Shopping Cart System

This is a console-based Shopping Cart System written in C#, submitted as a quiz requirement for the Computer Programming 2 course.

AI Usage

As someone who struggles with coding, seeking help and guidance from AI assisted me in developing this system. At first, since I was not familiar with navigating GitHub and its functions, I asked about concepts like commits and how to upload files. I asked how to start the program and what does the structure looks like, also I requested sample codes for specific sections, such as computing remaining stock and understanding how int.TryParse works, since I had forgotten its function. Additionally, I used AI to check for errors in my code, helping for validations and logic, and it suggest improvements, which helped me identify and fix mistakes. However, I made sure to understand and interpret each part of the code myself. Lastly, I asked AI for guidance on creating a flowchart, as I had no prior knowledge of it.

Question that I asked/Prompts:

- What is the concept of commits in GitHub?
- How to upload files?
- How should I start the program and what does the structure looks like? (Pasted the starting point of the program.)
- How int.TryParse works?
- Can I have a sample codes for computing the remaining stock?
- Is there any error in my code?
- How to make a flowchart?
- What does each shapes represent?

PART 2 SHOPPING CART SYSTEM

AI USAGE: I used it as a guide to understand some parts, but I wrote and modified the code myself. I asked for several codes to add or fix any of the logic I have inside my program. I asked help for the errors I found hard to fix, it gave me codes and I pasted it but I made sure to know the function of every code

that has given and to understand it. Some of the requirements are hard for me to, I do not know what code I should type to supply a certain requirement and I asked an AI for help, but I made sure to understand and interpret each part of the code myself.

Questions that I asked/Prompts:

- what classes I need, help me create classes for Product, CartItem, and maybe Order?
- How to structure CartItem class that connects a product with quantity?
- (pasted a code) what is wrong with my code, why is it not applying the discount for the item I purchased?
- How to store past transactions. Can you help me make an Order class with receipt number and final total?
- I'm confused how to store items added to the cart using arrays. Can you help me create and manage a CartItem array?
- How do I properly display cart items so that it shows the index, product ID, name, and quantity all in one line?"
- What are the codes I must include for an order history?
- What does "var p" means and its function?
- I want to search products even if the user types partial words. How do I check if a product name contains the search key in a case-insensitive way?
    - How to add a rceipt?

Summary of Changes

The updated version of the program shows a significant improvement compared to the first part of quiz code. In the first version, the system was very simple and only allowed users to view products, add items to the cart, and compute the total at the end of the program with a basic discount logic. It did not include advanced features such as searching, category filtering, cart management options, or order tracking. In the updated version, the program was expanded into a full menu-driven shopping system with multiple functionalities such as viewing products, searching by name, filtering by category, viewing and managing the cart, and a complete checkout process with payment validation, discount computation, receipt generation, and change calculation. A new Order class was also added to store order history with receipt numbers and final totals, allowing users to review past transactions. Additionally, the cart system was improved to support removing items, updating quantities, and clearing the cart while properly adjusting stock levels. Input validation was also enhanced to prevent invalid entries, especially during payment and menu selection. Overall, the updated code transforms the original basic structure into a more complete and realistic e-commerce system with better functionality, user interaction, and data handling.